Software Design Description 500

Remote Graphical Interface Environment Overview

References

SDD000 | Practices and implementation strategies used |

Overview

Platforms | Apple, Android, Web |
Languages | Dart / Flutter |
Interfacing components | ESP32-S3 (Receiver Module) |
Connection | WiFi AP "TrailCamera" → WebSocket ws://192.168.4.1:80/stream |

Scope of Module

- Receive reconstructed grayscale frames from ESP32-S3 over WebSocket (binary messages)
- Display live frame stream from trail camera
- Settings manager: SF, BW, TX power, packet count (runtime RF parameter adjustment)
- Remote control prompts sent over LoRa via S3 relay
- Cross-platform availability (Flutter)

Current Status

Implemented

- Project scaffolding (Flutter, screens/models/services/widgets structure)
- `lora_packet.dart` — LoRa packet data model

TODO

- WebSocket client connecting to S3 AP
- Frame display — render grayscale frames from binary WS messages
- Settings UI (SF, BW, TX power)
- Motor control interface